

Multi-Agent Reinforcement Learning: A Review of Challenges and Applications

Lucia Canese, Gian Carlo Cardarilli, Luca Di Nunzio, Rocco Fazzolari, Daniele Giardino, Marco Re, Sergio Spanò

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Systematic review of single and multi-agent reinforcement learning (MARL), Centralized Training with Decentralized Execution (CTDE), value factorization, and communication networks.

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Comprehensive literature synthesis across MARL algorithmic benchmarks.

Key Numerical Findings & Empirical Proofs:

Proves that unconstrained message broadcasting scales exponentially as $O(M^2)$, causing coordination bottlenecks, non-stationarity, and communication channel congestion.

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:

Classical pre-LLM paradigm; agents restricted to low-dimensional vector action spaces; lacks epistemic uncertainty gating for natural language communication.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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